

Project Report

Author

Full Name: Sukhdev Pratap Singh

Roll Number: 23f1001361

Student Email: 23f1001361@ds.study.iitm.ac.in

I am a diploma level student at IITM BS. I am also final year student at MGAHV Wardha.

Description

This is a Placement Portal Application which is used to simplify the tedious process of Placements in colleges. It has three controllers: student, companies, Admin. Student can apply for the job drive which is organised by Companies through the portal. Both student and companies should have to register themselves at portal. Admin can manage Drives, Students, Companies. If a user violates the rules, the administrator can block their account.

My Approach:

First I started thinking about the entities and their relationship and attributes. I designed my templates after that and later I moved on to backend. I changed the html templates along with the backend.

Technologies used

Technology/Library	Purpose
Flask	core backend web framework.
SQLAlchemy	object Relational Mapper for SQLite database.
Jinja2	Template engine for rendering dynamic HTML pages.
SQLite	Lightweight database used for storing application data.
HTML	Used for structure of web pages.
CSS	Used for styling the web pages.

AI/LLM Declaration

I used ChatGPT(GPT-5) to assist in writing SQLAlchemy model definitions, for styling and backend debugging.

The extent of AI/LLM usage is around 15-20%. Most of the logic, debugging and integration were done manually.

Project Report

Database Schema / ER Diagram

Tables

User - stores account details for every system user

(id, username, email, password, role, is_blocked)

Student - stores additional details for users whose role is student

(id, name, department, resume, user_id)

Company - stores details for users whose role is company

(id, name, website, is_verified, user_id)

PlacementDrive - stores job placement drives created by companies

(id, name, job_title, eligibility, job_description, deadline, is_completed, company_id)

Application - stores the applications submitted by students for placement drives

(id, student_id, drive_id, status)

Relationships

One-to-One: User → Student

Each user with the role **student** has exactly one corresponding student profile.

One-to-One: User → Company

Each user with the role **company** has exactly one corresponding company profile.

One-to-Many: Company → PlacementDrive

A company can create multiple placement drives.

One-to-Many: Student → Application

A student can apply to multiple placement drives.

One-to-Many: PlacementDrive → Application

A placement drive can receive applications from many students.

Additional Constraint

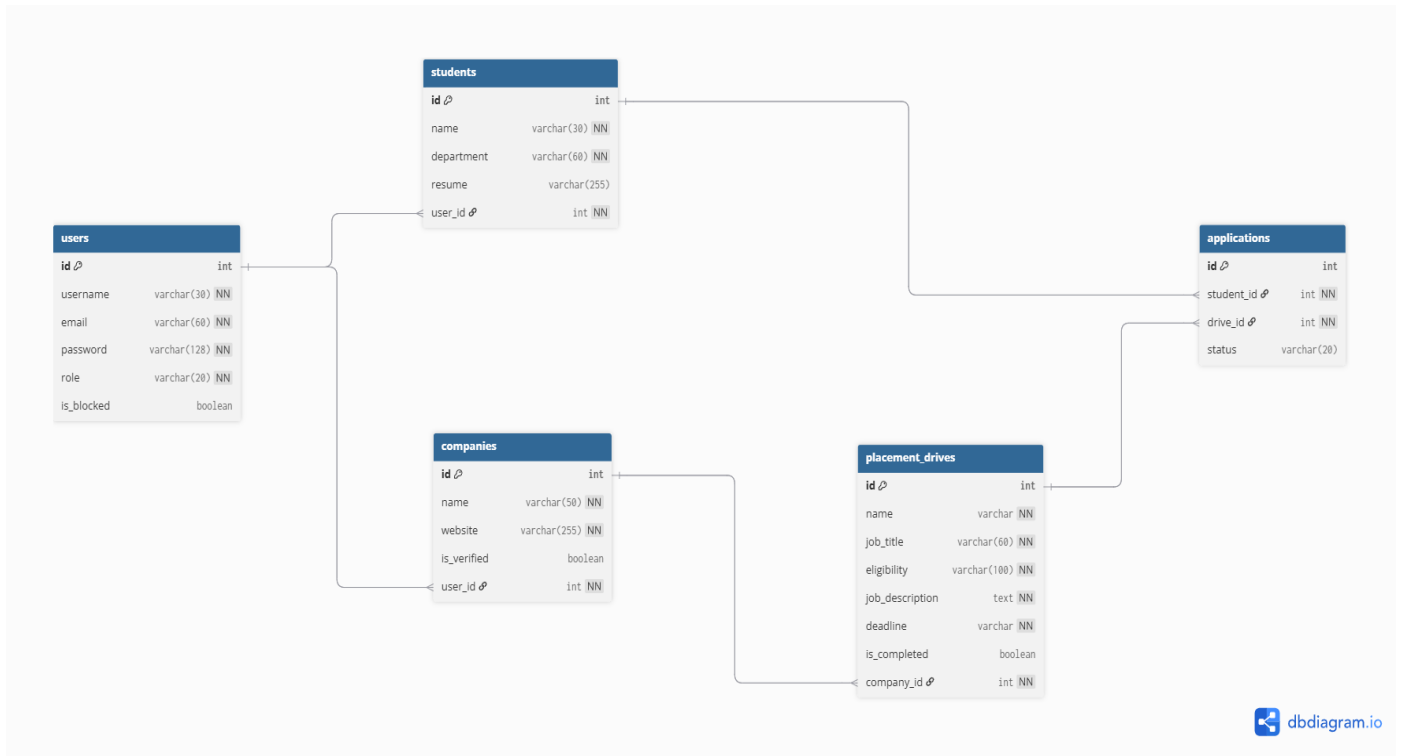
Unique Constraint: Application (student_id, drive_id)

This ensures that:

- A student **cannot apply to the same drive more than once**.

Project Report

ER-Diagram



Video Presentation

Drive Link: <https://drive.google.com/file/d/1Ckr2mJNSP1v8pwUDjpiUg2RBtLeIGkl9/view?usp=sharing>